# From Pixels to Perspectives: Audience-Specific Visual Sentiment in Political Imagery

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**Abstract**

Visual sentiment analysis typically assigns each image a single sentiment label, yet political images may elicit systematically different evaluations across audiences. We call the loss of this variation through aggregation *perspective collapse*. We develop a measurement framework that distinguishes shared evaluations from audience-specific interpretations and identifies when pooled labels recover a different quantity from the audience-specific evaluation of interest. Using 59,542 ratings of 960 immigration images from Democratic and Republican respondents, we show that partisan differences vary substantially across images and cannot be represented by a uniform audience adjustment. Models that preserve this heterogeneity recover audience-specific evaluations more accurately, with the largest gains for images that generate the greatest disagreement. Pooling can also change substantive conclusions about which visual features are associated with sentiment. These results show that reliable and predictable aggregate labels can still misrepresent the evaluations researchers seek to measure, making the intended audience a central part of computational measurement.

**Key words:** visual data, visual data labeling, sentiment, annotator disagreements, measurement, deep learning.

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# 1 Introduction

The rapid advance of visual social media platforms like Instagram, TikTok, and X has made images central to political communication where they direct public attention (Domke, Perlmutter and Spratt, 2002; Grabe and Bucy, 2014; Schill, 2012), shape attitudes and sentiments (Anastasopoulos et al., 2024; Bossetta and Schmøkel, 2023; Casas and Williams, 2019), and reinforce stereotypes (Carpinella and Bauer, 2021; Domke, Perlmutter and Spratt, 2002). These effects are amplified during pivotal moments like elections (Grabe and Bucy, 2014) or periods of heightened issue salience (de Lima-Santos et al., 2023; Casas and Williams, 2019), when the way events are visually depicted can shape public reaction. As political scientists increasingly use automated computer-vision methods to analyze large image corpora (Won, Steinert-Threlkeld and Joo, 2017; Williams, Casas and Wilkerson, 2020; Boussalis et al., 2021; Coan et al., 2021; Torres and Cantú, 2022; Peng and Lu, 2023; Torres, 2024; Boussalis, Coan and Holman, 2024; Arnold and Küpfer, 2024), they must convert human evaluations of political imagery into measurable labels. These labels rarely stay only in the measurement stage. Image sentiment enters as an explanatory variable when researchers ask whether the emotional tone of protest or campaign imagery predicts sharing and mobilization or voter reactions, as a treatment when experiments vary the tone of news photographs to move political attitudes, and as an outcome when studies characterize the tone of the imagery that outlets, candidates, or legislators choose to circulate. In each use the label reflects how some audience experiences the image, and creates the grounds for the substantive claims that follow. Ensuring that these labels correctly represent the evaluations researchers intend to study has thus become an important measurement problem.

Most visual-sentiment workflows collect evaluations from multiple annotators and aggregate them into a single image-level label, which is then treated as the target for manual or automated analysis. This practice is well justified when annotators provide imperfect measurements of a shared image-level construct. Annotators may differ in re-

liability or in how they use the response scale, but these differences can be modeled as properties of the measurement process, and additional annotations improve estimation of the underlying image-level quantity. Recent methodological work similarly considers how annotations should be aggregated when individual annotations are not independent, including in settings involving LLM-generated labels (Bouyamourn and Spirling, 2026). For subjective political content, however, viewer characteristics may affect the evaluation elicited by a particular image rather than merely the precision or scale on which it is reported. The same image of a protest, a border crossing, or a partisan political figure may be interpreted differently by viewers with different prior attitudes and political predispositions (Williams et al., 2026), and the magnitude of these differences can vary across images. In such settings, disagreement may reflect image-specific audience heterogeneity rather than random error around one common value.

Existing research in political measurement has long recognized that annotators can differ systematically. Latent-variable and item-response models can account for differences in annotator reliability, response thresholds, and use of the scale while retaining a common latent parameter for each case (Marquardt and Pemstein, 2018). Such models are appropriate when annotators use the scale differently but evaluate the images in broadly the same way. However, they might be inadequate when different audiences evaluate particular images differently. The relevant distinction is therefore not between simple aggregation and modeling annotator heterogeneity, but between a common-latent measurement model and an audience-dependent one.

This possibility is the central concern of the perspectivist turn in computational linguistics, where scholars argue that annotator identities, experiences, and beliefs can systematically shape labels assigned to subjective content (Davani, Díaz and Prabhakaran, 2022; Sap et al., 2022; Cabitza, Campagner and Basile, 2023; Van der Velden et al., 2025). Yet visual sentiment workflows generally continue to construct a single aggregate label for each image and use that label as the target of automated prediction (Borth et al., 2013;

You, Jin and Luo, 2017; Song et al., 2018; Ortis, Farinella and Battiato, 2020). The problem is not simply that aggregation removes disagreement. When evaluations vary systematically across audiences, a single label represents a particular mixture of those evaluations. We refer to the reduction of distinct audience-specific evaluations to a single aggregate label as *perspective collapse*. A single aggregate label may be appropriate when the mixture itself is the scientific target, but it should not automatically be interpreted as an audience-invariant property of the image.

Increasing the size or diversity of the annotation pool does not by itself resolve this measurement problem. Larger panels reduce individual-level error (Benoit et al., 2016; Williams et al., 2026), and more diverse panels can improve the representation of relevant audiences. However, if their evaluations are still collapsed into a single label, the resulting score will depend on the composition of the annotator pool. When audience differences vary across images, additional annotations improve the precision of the aggregate but do not ensure that it corresponds to the researcher's target population. We thus develop a target-aware procedure for determining when an aggregated score is adequate, when audience-dependent image parameters are needed, and how those parameters should be aggregated for a declared target population.

Our argument has two parts, addressed to two different readers. The first is for researchers who care about how a particular audience evaluates political imagery, or about the differences between audiences. For them the implication is direct: when the audience gap varies across images, no single label with a uniform adjustment can represent the evaluations of interest, and the measurement output should retain separate audience-specific estimates. The second part is for the group of researchers who want, and often need, one sentiment score per image. For them the question is not whether to pool but what a pooled score measures. A pooled label estimates a mixture of audience evaluations weighted by whoever happened to annotate each image. When those weights differ from the population the researcher has in mind, or vary from image to image, the pooled

label is a moving target, and adding annotators does not fix it. The procedure we develop gives this reader a way to check whether a pooled label is an adequate target at all, which it is when audiences agree or differ by a constant; to construct it with declared rather than incidental weights when it is not; to improve on the raw average through model-based estimation; and to tell whether an association estimated on the pooled label is shared across audiences or concentrated in one. The two claims are closely related, but they matter to different readers.

We apply this procedure to 960 immigration-related images evaluated by Democratic and Republican respondents. We first examine whether partisan disagreement reflects only a general difference between the two audiences or whether the size of that difference varies systematically across images. We show that substantial image-specific partisan heterogeneity remains after accounting for respondents' general immigration attitudes and stable respondent-level response tendencies. We then compare an audience-invariant measurement model with an audience-dependent hierarchical model. Finally, we use machine-learning models to compare recovery of observed Democratic, Republican, and pooled image-level evaluations from visual content. We show that the consequences of pooling are concentrated among images that generate the greatest audience disagreement and can alter conclusions in downstream analyses that rely on these labels.

This study shows that the relevant measurement question is whether annotators' evaluations can be represented by a single latent image parameter, which is a different question from whether they disagree at all. When image evaluations vary systematically across audiences, recovering one pooled score can mask substantively meaningful variation even when the score is estimated precisely. Our argument is most directly relevant to labels that characterize how visual content is evaluated or experienced by an audience, rather than labels that simply identify what is depicted in an image. Whether an image contains a flag, a protester, or a smiling face may often be treated as a property of its content; whether that same image conveys a positive or negative portrayal may de-

pend systematically on the audience evaluating it. This concern also connects to broader measurement literatures on item response theory, differential item functioning, and hierarchical measurement models, which provide tools for modeling systematic variation in observed responses across raters or groups. Our framework extends existing approaches to annotator heterogeneity by making explicit whose evaluations the resulting measure is intended to represent. Although we examine this phenomenon in the context of partisan evaluations of immigration imagery, the same problem arises whenever computational labels aggregate subjective evaluations that may differ systematically across audiences.

## 2 Perspective-Dependent Visual Sentiment

Traditional approaches to visual sentiment analysis generally assign each image a single sentiment score that can be estimated through human annotation or automated classification (Borth et al., 2013; Ortis, Farinella and Battiato, 2020; Zhao et al., 2021). Although these approaches differ in methodology, they typically assume that additional annotations or improved prediction make it possible to estimate that score more accurately. In such a framework, disagreement among annotators is treated primarily as measurement error or variation in how individuals use the response scale, while aggregation is used to produce a more stable image-level estimate.

The assumption of a single recoverable sentiment becomes less convincing when applied to political imagery. Appraisal theories of emotion hold that affective responses depend on the interaction between external stimuli and observers' goals, beliefs, and prior experiences (Lazarus, 1991; Scherer, 2001). Political predispositions can therefore shape how individuals perceive and evaluate the same information (Kunda, 1990; Leong et al., 2019). Consistent with this account, identical immigration images can evoke different reactions across partisan audiences (Madrigal and Soroka, 2023; Williams et al., 2026). Perspectivist approaches to annotation similarly treat systematic disagreement as infor-

mation about audience-specific evaluations of subjective content (Sap et al., 2022; Davani, Díaz and Prabhakaran, 2022; Cabitza, Campagner and Basile, 2023). Existing evidence that audiences differ, however, does not establish whether those differences can be represented by a uniform audience adjustment or whether they vary across images. The latter possibility would imply that sentiment cannot be represented by a single image-level parameter shared across audiences.

These differences raise two related measurement questions. First, can annotators' evaluations be represented by a single image-level parameter, or does the evaluation elicited by an image vary across audiences? Existing measurement models can accommodate differences in annotator reliability, response thresholds, and use of the measurement scale while retaining one parameter for each image (Marquardt and Pemstein, 2018). They are less suitable for settings in which the size or direction of audience differences varies across images. Second, when distinct audience-specific evaluations are combined, the resulting aggregate represents a particular mixture of those audiences. Whether such an aggregation is appropriate depends on the population whose perceptions the researcher intends to measure.

Standard agreement and reliability diagnostics (such as Cohen's $\kappa$ and Krippendorff's $\alpha$) summarize the extent of annotator disagreement but do not identify its source.[1] In particular, they cannot distinguish random measurement error from systematic image-by-audience heterogeneity. Such diagnostics are useful for assessing the consistency of image-level labels, but insufficient for determining whether annotators' ratings can be represented by a single image-level parameter.

As a result, we distinguish between two sources of annotator disagreement. In the first, annotators provide noisy measurements of the same image-level evaluation. In the

[1]Low reliability is often treated as evidence that coding instructions should be clarified, ambiguous cases resolved, or a majority rule imposed to define a single label (Sabou et al., 2014; Yan et al., 2014; Äyräväinen, Hinds and Davidson, 2025). These steps may be appropriate when disagreement reflects confusion or noise, but can remove theoretically meaningful variation when disagreement is structured by audience.

second, the evaluation elicited by an image varies systematically across audiences. When these audience-specific evaluations are combined into a single label, the resulting aggregation removes information about the distinct perspectives represented in the data (*perspective collapse*). We now formalize the distinction between those two cases.

Let $Y_{ijg}$ denote the sentiment rating assigned to an image $i$ by annotator $j$ belonging to audience group $g$. To make this distinction precise, consider first a shared-evaluation model in which each observed rating is a noisy measurement of a single image-level evaluation,

$$Y_{ijg} = \theta_i + \varepsilon_{ijg} \tag{1}$$

where $\theta_i$ denotes the shared image-level evaluation and $\varepsilon_{ijg}$ is an unobserved, stochastic annotator-level error. If the errors are independent with mean zero, $\mathbb{E}[\varepsilon_{ijg}] = 0$, disagreement reflects random measurement error around $\theta_i$. Aggregation then averages away idiosyncratic deviations, and the consensus estimator converges to $\theta_i$ as the number of annotations increases.

When disagreement reflects systematic differences in interpretation rather than random deviations around a shared image-level evaluation, observed ratings should instead be represented as

$$Y_{ijg} = \theta_{ig} + \varepsilon_{ijg}, \tag{2}$$

where $\theta_{ig}$ denotes the audience-specific image-level evaluation. For two audience groups, let $\Delta_i = \theta_{iA} - \theta_{iB}$ denote the audience difference for image $i$. This difference can be decomposed as $\Delta_i = \beta + \delta_i$, where $\beta$ captures the average tendency for one audience to evaluate images more positively or negatively than the other and $\delta_i$ is the image-specific deviation from that average. A constant audience shift is represented by $\delta_i = 0$ for all images, whereas image-specific audience heterogeneity arises when $\mathrm{Var}(\delta_i) > 0$, so that the size as well as potentially the direction of the audience difference varies across images.

Under this model, the pooled label estimates a mixture of audience-specific evalua-

tions. Let $n_{ig}$ denote the number of annotations contributed by audience group $g$ to image $i$, let $n_i = \sum_g n_{ig}$, and define the realized annotation weight $\pi_{ig} = n_{ig}/n_i$. The pooled label can be written as

$$\bar{Y}_i = \sum_g \pi_{ig} \bar{Y}_{ig}. \tag{3}$$

Conditional on the realized allocation of annotators across audience groups,

$$\mathbb{E}\left[\bar{Y}_i \mid n_{i1}, \ldots, n_{iG}\right] = \sum_g \pi_{ig} \theta_{ig} \equiv \theta_i(\pi_i). \tag{4}$$

As the number of annotations increases while the relative audience composition remains stable, the pooled label converges to this annotation-weighted quantity. This quantity is not inherently invalid, but its meaning depends on the composition of the annotation pool. Moreover, when $\pi_{ig}$ varies across images, the audience mixture represented by the pooled label also varies across the image corpus.

To determine whether such an annotation-weighted quantity is appropriate, we must distinguish the composition of the annotation pool from the population whose perceptions we intend to measure. Let $q_g$ denote the researcher-specified weight of audience group $g$ in the population whose evaluations the study intends to measure, where $\sum_g q_g = 1$. Unlike the annotation weights $\pi_{ig}$, the target weights $q_g$ represent a substantive design choice and are held constant across images. The intended target is

$$\theta_i(q) = \sum_g q_g \theta_{ig}. \tag{5}$$

The difference between the annotation-weighted quantity and the intended target is

$$B_i(\pi_i, q) = \theta_i(\pi_i) - \theta_i(q) = \sum_g (\pi_{ig} - q_g)\theta_{ig}. \tag{6}$$

For two audience groups, using $\Delta_i = \theta_{iA} - \theta_{iB}$, this discrepancy reduces to

$$B_i(\pi_i, q) = (\pi_{iA} - q_A)\Delta_i. \tag{7}$$

The pooled label recovers the researcher's declared target when the observed annotation weights match the target weights, $\pi_i = q$, or when $\Delta_i = 0$, in which case audience composition becomes irrelevant. Otherwise, increasing the number of annotations reduces sampling error around $\theta_i(\pi_i)$ but does not eliminate its discrepancy from $\theta_i(q)$. Quota sampling or post-stratification can align $\pi_i$ with $q$ for a specified target population, but only after that target population has been substantively defined.

Even when the observed annotation weights match the target weights, aggregation replaces the separate audience-specific evaluations with a single weighted average. This is appropriate when that weighted average is itself the quantity the researcher intends to study. It is insufficient, however, when the research question concerns a particular audience, compares audiences, or treats sentiment as an innate property of the image.

The consequences of aggregation also depend on the degree of audience disagreement for a given image. When $|\Delta_i|$ is small, the pooled quantity may closely approximate both audience-specific evaluations. As $|\Delta_i|$ increases, however, no single mixture can represent both audiences equally well, making the choice of target and weighting rule more consequential. The costs of aggregation are expected to be concentrated among images that elicit the greatest audience disagreement.

Three cases follow from this formulation. First, when $\Delta_i = 0$, audiences share the same image-level evaluation, and a single label is sufficient. Second, when $\Delta_i = \beta$, audiences differ by a constant amount but retain the same relative image structure, so a common image parameter with an audience adjustment may be sufficient. Third, when $\Delta_i = \beta + \delta_i$ and $\mathrm{Var}(\delta_i) > 0$, the audience difference varies across images, and no single image score with a constant audience correction can represent both sets of evaluations. The last case is where the model must allow image-level evaluations to vary by audience.

This formalization yields three observable implications for annotation and measure-

ment. Together, they specify when a pooled label is appropriate and when preserving image-by-audience variation and audience-specific labels should improve measurement.

**Implication 1.** *Information loss from aggregation increases with systematic audience disagreement.* When audience-specific evaluations are close, a pooled label can approximate both. As $|\Delta_i|$ increases, however, the pooled quantity moves further from the audience-specific evaluations, even when it is estimated precisely.

**Implication 2.** *Aggregate measures of performance can mask where information loss occurs.* Pooling does not necessarily distort the entire image corpus uniformly. Images evaluated similarly across audiences remain reasonably represented by one label, whereas images that generate the largest audience differences account for most of the measurement loss.

**Implication 3.** *Audience-specific structure should emerge for substantively meaningful and reproducible audience partitions, not arbitrary splits.* A theoretically motivated audience characteristic should generate image-by-audience differences that exceed those produced by random partitioning and can be measured reliably across annotators.

This framework implies a concrete measurement procedure. Researchers should first define the audience, or mixture of audiences, whose evaluations constitute the target and retain the individual annotations needed to evaluate audience heterogeneity. Candidate audience distinctions should be motivated by theory or prior evidence rather than selected solely because they generate disagreement. Researchers should then compare a model with no audience difference, a model with one uniform audience adjustment, and a model in which the audience difference varies across images. The image-varying structure should be retained when it improves recovery of audience evaluations among unseen respondents and the audience-specific image estimates can be measured with sufficient

reliability. If the data support no audience difference, a pooled image label is sufficient. If they support only a uniform difference, a single audience adjustment can be used. If the audience difference varies across images, the measurement output should instead retain separate model-based evaluations for each audience and image. These estimates should be aggregated only when the research question calls for a single measure across audiences, using explicitly specified weights $q$. When the available annotations are too sparse to distinguish reliably among these cases, researchers should report that uncertainty rather than simply defaulting to a pooled label.

## 2.1 Estimator Choice Under Audience Heterogeneity

We use simulations to examine how estimator performance depends jointly on the structure of audience disagreement and the target the researcher seeks to recover. We generate ratings from two audiences under three conditions: no systematic audience difference, a uniform difference across images, and an audience difference that varies by image. We compare a raw pooled mean,[2] an estimator with a single audience adjustment, and a hierarchical model that allows the audience difference to vary across images.

We evaluate each model by its root-mean-squared error (RMSE) in recovering the Audience B image-level target, $q = (0, 1)$. Holding the target and the underlying random-effects structure constant isolates the consequences of imposing different assumptions about audience disagreement. A model with no audience difference should suffice when no gap exists, a uniform adjustment should suffice when the gap is constant across images, and an image-varying model should perform best when the audience difference itself varies by image.

Table 1 shows that model performance depends on whether its audience structure matches the data-generating process. When no audience gap exists, all three models per-

[2]The simulation uses the arithmetic mean as the conventional pooled estimator. Other aggregation rules can include inverse-variance-weighted means and, for categorical annotations, majority vote. These rules construct the aggregate differently but still replace audience-specific evaluations with a single quantity.

**Table 1: Recovery of an audience-specific target under simulated audience structures**

| Model | No gap | Uniform gap | Image-specific heterogeneity |
|---|---|---|---|
| No audience difference | 0.187 | 0.272 | 0.439 |
| Uniform audience difference | 0.193 | 0.196 | 0.395 |
| Image-varying audience difference | 0.195 | 0.198 | 0.261 |

*Note:* The reported values indicate RMSE, in points on the 1–7 scale, in recovering the Audience B image-level target, $q = (0, 1)$, across 300 Monte Carlo replications. Each replication contains 80 images and 30 ratings per audience per image. Parameters are calibrated to the scale and sparsity of the immigration application, with $\mathrm{SD}(\delta_i) = 0.65$ in the image-specific condition. All models include respondent and image random intercepts and differ only in whether they permit no audience difference, one uniform audience difference, or an image-varying audience difference. Lower values indicate more accurate recovery. Results for the equally weighted target, raw-mean estimators, the attitude-by-image scenario, and interval coverage appear in Table A.1 of the Appendix. Because every model is evaluated on the same simulated datasets, differences are assessed by paired comparisons within replication. The Monte Carlo standard error is at most 0.003 for every reported RMSE and for every paired difference, so all differences discussed exceed simulation noise.

form similarly. Under a uniform gap, the uniform and image-varying models yield nearly identical RMSE and both substantially outperform the no-difference model. When the audience gap varies across images, however, the image-varying model reduces RMSE from 0.395 under the uniform model to 0.261. Thus, additional audience-specific flexibility provides little benefit when a simpler structure is adequate, but substantially improves recovery when disagreement is image-specific.

# 3 Data

Empirically testing this framework requires data that preserve individual sentiment ratings. We constructed such a dataset using politically salient immigration imagery labeled separately by Democratic and Republican respondents.

## 3.1 Image Corpus

We chose immigration as our empirical setting for two reasons. First, Democratic and Republican attitudes toward immigration differ systematically (Card et al., 2022; Ollerenshaw and Jardina, 2023), providing a theoretically motivated basis for expecting audience-specific evaluations. Second, immigration is represented visually in highly varied ways,

including images emphasizing border enforcement, detention, protests, migrant crowds, and families or children. Prior work shows that how immigration is visually represented can shape audience responses (Gasparyan and Sirotkina, 2025). This variation makes immigration especially useful for examining whether partisan differences are adequately represented by a uniform audience adjustment or instead vary across images.

Many immigration images do not contain a single, unambiguous emotional meaning. Unlike images with commonly recognizable affective cues, such as smiling faces, natural disasters, or graphic violence, immigration imagery is often interpreted in light of how viewers understand the broader social and political context. The same image of migrants walking along a road, waiting at a border crossing, or traveling with family members may evoke compassion, concern, or indifference depending on the observer's prior beliefs about immigration.

We assembled a corpus of immigration-related images from publicly available posts by major U.S. news organizations on X (formerly Twitter) and supplemented this collection with images obtained from Getty Images, a primary source of visual content used in news reporting on immigration. After removing duplicate and near-duplicate images, the final corpus consists of 960 unique images spanning a diverse range of immigration-related events, settings, and visual frames.

Our corpus is not intended to constitute a probability sample of all immigration imagery. These images were selected for dissemination by major news organizations and reflect upstream processes of news production, editorial selection, and newsworthiness. These processes do not by themselves invalidate the within-image comparison at the center of our analysis—whether different audiences systematically evaluate the same image differently—but they do affect corpus-level quantities such as the average partisan gap, the prevalence of high-disagreement images, and the average cost of pooling. We therefore interpret these quantities as properties of the observed corpus rather than population estimates for all immigration imagery.

## 3.2 Sentiment Annotations and Label Construction

We collected sentiment evaluations from 5,533 unique respondents who self-identified as either Democrats or Republicans. Participants were randomly assigned a subset of images and rated each image on a seven-point sentiment scale.[3] On average, each image was evaluated by approximately 30 Democratic and 30 Republican respondents.

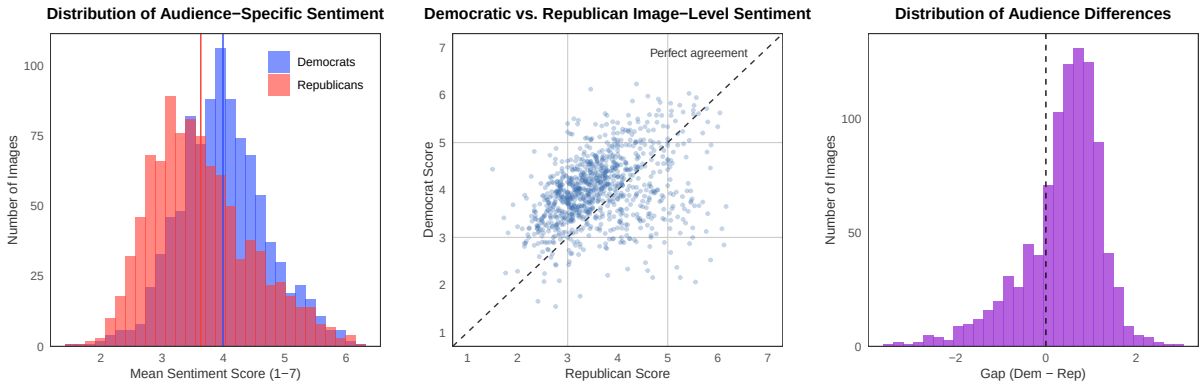
For each image, we compute separate sentiment scores by averaging ratings within each partisan group, along with a pooled score that averages ratings across all respondents irrespective of their partisanship. The audience-specific means summarize each audience's average evaluation of the image, whereas the pooled score collapses those evaluations into a single image-level label. This pooled score corresponds to the conventional consensus-label approach in visual sentiment analysis.

We first examine the extent to which Democratic and Republican annotators differ in their evaluations of the same images. If visual sentiment is perspective-dependent, we should observe systematic divergence between the two audience-specific labels rather than only random differences around a common image-level score. With approximately 30 ratings per audience for each image, the resulting audience-specific image means are estimated with reasonably high reliability (Democratic ICC$(1, k) = 0.79$; Republican ICC$(1, k) = 0.84$; see Table B.1 in the Appendix).

Figure 1 shows considerable overlap in the distributions of Democratic and Republican image-level sentiment evaluations (left panel), but Democratic respondents assign more positive ratings on average. Across the 960 immigration images, the mean Democratic rating exceeds the mean Republican rating by 0.36 points on the 7-point sentiment scale. The middle panel shows that the size of the audience difference varies widely across images. While many images lie close to the line of perfect agreement, others gen-

[3]We asked respondents the following question: "Would you say that this image portrays the subject(s) or object(s) in this picture in a positive or negative light? '1' stands for negative, '4' is neutral, and '7' stands for positive." Our outcome measures perceived portrayal—the valence respondents attribute to how the image depicts its subjects or objects—rather than the respondent's own emotional reaction to the image.

## Figure 1: Audience-specific sentiment evaluations




*Note:* $N = 960$ immigration images; each image's score is the mean of individual ratings within an audience (Democratic or Republican) on a 1–7 sentiment scale (1 = very negative, 4 = neutral, 7 = very positive). Each observation corresponds to the same immigration image evaluated independently by Democratic and Republican respondents. The left panel shows the distribution of audience-specific image means, with vertical lines at the Democratic and Republican means; the middle panel plots Democratic against Republican image means (one point = one image; dashed 45° line = perfect agreement; faint guides mark the neutral-band edges at 3 and 5); and the right panel shows the per-image gap (Democratic minus Republican; dashed line at zero). Mean sentiment is 3.99 ($SD = 0.72$) among Democratic respondents and 3.63 ($SD = 0.83$) among Republican respondents, producing a mean Democratic–Republican gap of 0.36 points (95% CI $[0.30, 0.42]$; paired $t(959) = 12.6$, $p < .001$; Cohen's $d = 0.41$, 95% CI $[0.34, 0.47]$). The two image-level scores are positively but imperfectly correlated ($r = 0.35$, 95% CI $[0.29, 0.40]$), and Democratic respondents rate 74.5% of images (715 of 960) more positively than Republican respondents. These intervals reflect sampling error in the per-image partisan ratings and treat the 960 images as fixed; they do not represent sampling of images from a larger population, consistent with the corpus caveat. Because each image mean rests on roughly 30 ratings, the raw dispersion of these gaps is itself inflated by sampling error; we separate that noise from systematic variation with the permutation benchmark in Table 2. The model-based counterpart of this descriptive gap, which adjusts for respondent and image baselines, is reported in Table 2 in the opposite orientation (Republican minus Democratic) as $\beta = -0.352$.

erate large disagreement between Democratic and Republican respondents. The right panel shows that the distribution of Democratic–Republican gaps is shifted to the right, indicating that Democratic respondents assign higher sentiment scores to most images.

Figure 1 demonstrates both an average partisan difference and wide image-to-image variation in the size of that difference. Descriptive variation alone, however, cannot establish whether the audience gap itself varies systematically across images or whether the observed spread reflects sampling error around a uniform audience difference. To distinguish these possibilities, we fit a crossed random-effects model to the 59,542 individual immigration-image ratings. The model decomposes the audience gap for image $i$ as $\theta_{iR} - \theta_{iD} = \beta + \delta_i$, where $\beta$ is the average Republican–Democratic difference and $\delta_i$ is the image-specific deviation from that average. The model also accounts for respondent-level response tendencies and differences in image baselines. Table 2 reports the resulting estimates and the respondent-level permutation test.

**Table 2: Variation in Republican–Democratic evaluation gaps across images**

| Quantity | Estimate | 95% interval |
|---|---|---|
| *Model-based estimates* | | |
| Average Republican–Democratic difference, $\beta$ | $-0.352$ | $[-0.425,\ -0.280]$ |
| SD of image-specific deviation, $\mathrm{SD}(\delta_i)$ | $0.659$ | $[0.618,\ 0.702]$ |
| $\mathrm{SD}(\delta_i)/\lvert\beta\rvert$ | $1.87$ | $[1.58,\ 2.32]$ |
| *Distribution of estimated image gaps* | | |
| Gap larger in magnitude than $\lvert\beta\rvert$ | 72% | — |
| Gap reverses the average direction | 19% | — |
| *Respondent-level permutation test* | | |
| Observed variance of raw image gaps | $0.791$ | — |
| Mean variance under permutation | $0.287$ | — |
| Excess variance above permutation benchmark | $0.504$ | — |
| Share above permutation benchmark | 64% | — |
| Permutation $p$-value ($B = 2{,}000$) | $0.0005$ | — |
| *Supporting likelihood-ratio test* | | |
| Likelihood-ratio $\chi^2$ | 1368 | — |
| Approximate $p$-value | $< 0.001$ | — |

*Note:* Results use 59,542 ratings of 960 immigration images from 5,533 respondents. The random-effects model is $\mathrm{y}_{ij} = \mu + \beta\,\mathrm{Republican}_j + \delta_i\,\mathrm{Republican}_j + b_j + a_i + \varepsilon_{ij}$, where $\beta$ is the average Republican–Democratic difference and $\delta_i$ is the image-specific deviation from that difference. Bracketed quantities are 95% confidence intervals: for $\beta$, the Wald interval from the fitted model; for $\mathrm{SD}(\delta_i)$ and the ratio $\mathrm{SD}(\delta_i)/|\beta|$, percentile intervals from a parametric bootstrap (200 replications). The two gap-distribution shares summarize the shrunken empirical-Bayes point estimates $\hat{\beta} + \hat{\delta}_i$ and are reported without an interval; the permutation and likelihood-ratio rows are test diagnostics, for which no interval applies. For the respondent-level permutation test, party labels are reassigned across respondents while each respondent's set of sentiment ratings is preserved; the permutation distribution represents the image-gap variation expected under the null that party is unrelated to the pattern of ratings, and the $p$-value is reported with $B = 2{,}000$ permutations. The likelihood-ratio test compares the baseline model with one constraining $\delta_i = 0$.

The estimated average difference is $\beta = -0.352$ $[-0.425, -0.280]$ on the 1–7 scale, indicating that Republican respondents rate immigration imagery more negatively than Democratic respondents on average. The estimated image-specific variation is considerably larger: $\mathrm{SD}(\delta_i) = 0.659$ $[0.618, 0.702]$, approximately 1.87 times $[1.58, 2.32]$ the magnitude of the average difference. The estimated audience gaps therefore vary widely across images rather than clustering around a uniform shift. For 72% of images, the gap is larger in magnitude than $|\beta|$, and for 19% it reverses direction, with Republican respondents evaluating the image more positively than Democratic respondents.

To assess whether the image-to-image heterogeneity exceeds what would be expected if party were unrelated to the pattern of ratings, we conduct a respondent-level permutation test. Because party is a respondent-level characteristic, we shuffle party assignments across respondents while preserving each respondent's complete set of ratings and the repeated-measures structure. The observed variance of the raw image-level audience gaps is 0.79, compared with 0.29 under the permutation distribution. The exchangeability null accounts for only 36% of the observed gap variance; the remaining 64% is excess variation ($p_{\mathrm{perm}} < 0.001$). The corresponding estimate of $\mathrm{SD}(\delta_i) = 0.71$ is close to the model-based estimate of 0.66. A likelihood-ratio test comparing a model with a uniform audience difference to one allowing image-specific audience differences yields the same conclusion ($\chi^2(2) = 1368$, $p < 0.001$). These results indicate that the audience difference varies systematically across images rather than being adequately represented by a uniform adjustment.

Because partisanship may partly capture broader immigration attitudes, Table C.1 re-estimates the model with an attitude to immigration as a control. Doing so reduces both the average partisan difference and its image-specific variation, but substantial image-specific heterogeneity remains.

The purpose of these comparisons is to determine what image-level quantity should subsequently serve as the target of computational measurement: a single common label, a

common label with a uniform audience adjustment, or separate audience-specific labels. The variance decomposition establishes that the Republican–Democratic difference varies across images. We next assess whether retaining this image-specific variation improves measurement relative to models that impose a more restrictive audience structure. We compare three models with respondent and image random effects: a model with no audience difference, a model that permits one uniform Republican–Democratic difference across images, and a model that allows the audience difference to vary by image. Table 3 compares their overall fit, out-of-sample prediction of individual ratings, and recovery of audience-specific image means among unseen respondents.

**Table 3: Comparison of uniform and image-varying audience-difference models**

*(a) Model fit and out-of-sample prediction of individual ratings*

| Model | ΔBIC | CV RMSE |
|---|---|---|
| No audience difference | 1466 | 1.784 |
| Uniform audience difference | 1346 | 1.773 |
| Image-varying audience difference | 0 | 1.752 |

*(b) Audience-mean recovery among unseen respondents*

| Disagreement | Images | Pooled (raw) | Pooled (model) | Raw means | Uniform difference | Image-varying difference | Gain over uniform |
|---|---|---|---|---|---|---|---|
| Low | 308 | 0.756 | 0.726 | 0.748 | 0.689 | 0.691 | 0% $[-1, 1]$ |
| Mid | 308 | 0.707 | 0.676 | 0.663 | 0.646 | 0.615 | 5% $[3, 7]$ |
| High | 308 | 0.896 | 0.858 | 0.881 | 0.853 | 0.753 | 12% $[9, 15]$ |
| Overall | 924 | 0.790 | 0.757 | 0.770 | 0.735 | 0.689 | 6% $[5, 8]$ |

*Note:* Results use 59,542 immigration-image ratings from 5,533 respondents across 960 images. All models include random effects for individual response tendencies and image baselines. The uniform-difference model includes one average partisan difference, $\beta$, while constraining that difference to be the same across images. The image-varying model adds an image-specific random slope for party, producing audience-specific image estimates. ΔBIC is measured relative to the best-fitting model. Panel (a) reports five-fold cross-validation on unseen respondents; lower RMSE indicates better predictive performance. In panel (b), respondents are assigned within party to separate estimation and evaluation samples. Disagreement terciles are defined using only the estimation sample. Estimates from the estimation sample are then evaluated against Democratic and Republican image means from the unseen evaluation sample. The two "Pooled" columns apply a single image label, with no audience distinction, to both audiences: "Pooled (raw)" is a basic mean across all estimation-sample respondents (the naive aggregated label), and "Pooled (model)" is the model-based no-audience-difference score. The "Raw means" column uses unpooled audience means from the estimation sample. The recovery analysis includes 924 images with at least three ratings per party in *both* the estimation and evaluation half-samples—fewer than the 960 images used where the threshold is applied to the full sample (e.g., Table 5), because the split imposes the requirement within each half. Gain is the percentage reduction in RMSE from the uniform-difference model to the image-varying model. In panel (a), the cross-validated RMSE difference between the uniform and image-varying models is assessed by a paired test on the held-out ratings. In panel (b), bracketed intervals on the gain are 95% confidence intervals from an image-level bootstrap ($B = 2{,}000$) that resamples images within each tercile.

Panel (a) shows that allowing the audience difference to vary by image substantially

improves overall model fit. BIC is 1,346 points higher for the uniform-difference model than for the image-varying model, and the model with no audience difference performs still worse. The image-varying model also better predicts individual ratings from unseen respondents, reducing cross-validated RMSE from 1.773 to 1.752. This improvement is statistically reliable despite its modest size: on the same held-out ratings, the image-varying model's mean squared error is lower than the uniform model's by 0.076 ($[0.059, 0.093]$, with standard errors two-way clustered by respondent and image). The improvement in individual-rating RMSE is modest because individual ratings contain considerable respondent-level variation, but it is consistent with the fit statistics in favoring a model that retains image-specific audience differences.

Panel (b) provides a more direct test of the image-level quantities the models are intended to recover. Respondents are assigned within party to separate estimation and evaluation samples, and disagreement terciles are defined using only the estimation sample. Estimates from the estimation sample are then evaluated against Democratic and Republican image means from the unseen evaluation sample. When disagreement is low, the uniform and image-varying audience-specific models perform nearly identically (RMSE $= 0.689$ and $0.691$; gain $0\%$ $[-1\%, 1\%]$). As disagreement increases, however, retaining image-specific audience differences becomes more consequential. The image-varying model reduces RMSE relative to the uniform-difference model by 5% in the middle tercile $[3\%, 7\%]$ and by 12% in the highest tercile $[9\%, 15\%]$, from 0.853 to 0.753. Overall, it reduces audience-mean recovery error by 6% $[5\%, 8\%]$.

Panel (b) also separates two measurement choices. The first is whether to estimate a single pooled image evaluation or retain audience-specific evaluations. The second is whether to rely on raw averages or model-based estimates. Model-based estimates improve recovery relative to their raw counterparts: the model-based pooled estimate outperforms the raw pooled mean, while both audience-specific models outperform the raw audience means overall. More importantly, the advantage of retaining audience-specific

evaluations grows with disagreement. If the research question calls for a single measure across audiences, pooling remains an appropriate target, with model-based estimation providing a modest precision gain over simple aggregation. If the goal is to recover audience-specific evaluations, separate raw means already preserve that distinction, but model-based estimation improves their precision. Allowing the audience difference itself to vary across images provides the largest gains where disagreement is greatest. Having established that the empirical data support image-specific audience evaluations, we next examine what is lost when computational models are nevertheless trained on a single pooled target.

# 4 Computational Measurement

To examine how aggregation affects computational measurement, we intentionally hold the label-construction rule fixed. We train otherwise identical models using raw image-level means: separate Democratic and Republican means and a pooled mean across all respondents. Table 3 shows that hierarchical estimation recovers the underlying audience-specific quantities more precisely than raw means, but the purpose here is different. Using the same transparent aggregation rule for all three prediction targets isolates the consequence of preserving versus collapsing audience-specific evaluations before model training, rather than combining that distinction with differences in shrinkage or label estimation. The pooled raw mean also uses approximately twice as many ratings per image and is therefore the most reliable of the three raw targets (Table B.1), which if anything favors its predictability. Predictive performance itself, however, does not determine which target is substantively appropriate. Instead, we ask whether a model trained on the pooled labels can serve as a proxy for audience-specific evaluations and whether any resulting loss is concentrated among images with the largest audience differences.

We treat sentiment recovery as a continuous prediction task, predicting the image-

level mean sentiment score on the original 1–7 scale. All models use frozen CLIP ViT-B/32 image embeddings[4] with ridge regression. Model development uses 822 images with five-fold cross-validation, followed by final evaluation on a separate test set of 138 images. Holding the image representation, prediction head, and evaluation procedure fixed isolates differences across sentiment targets. Comparisons with alternative image representations and prediction architectures appear in Appendix D. All three targets contain substantial visually predictable variation: on the held-out test set of 138 images, the models explain 43% $[29\%, 54\%]$ of the variation in Democratic labels, 57% $[45\%, 65\%]$ in Republican labels, and 54% $[43\%, 63\%]$ in pooled labels (see Table D.1 in the Appendix).[5] Because these intervals overlap substantially, overall predictability does not distinguish the three targets and therefore does not identify which is substantively appropriate; the relevant comparison is whether pooled predictions adequately approximate audience-specific evaluations when the two audiences disagree.

**Table 4: Prediction accuracy by degree of audience disagreement**

| Disagreement | Gap range | $n$ | Audience-specific | Pooled proxy | Cost of pooling |
|---|---|---|---|---|---|
| Low | 0.00–0.48 | 46 | 0.441 | 0.404 | $-0.04$ $[-0.07, -0.01]$ |
| Medium | 0.49–0.98 | 46 | 0.353 | 0.442 | 0.09 $[0.04, 0.14]$ |
| High | 0.98–3.33 | 46 | 0.451 | 0.725 | 0.28 $[0.22, 0.34]$ |
| Overall | 0.00–3.33 | 138 | 0.415 | 0.524 | 0.11 $[0.08, 0.14]$ |

*Note:* Test images are grouped into terciles according to the absolute difference between their Democratic and Republican image-level sentiment evaluations. Table reports mean absolute prediction error (MAE), in points on the 1–7 scale, averaged across the two audiences. Audience-specific error compares each audience model with its corresponding audience label. Pooled-proxy error evaluates the pooled prediction separately against the Democratic and Republican labels and averages the resulting errors. Cost of pooling is pooled-proxy MAE minus audience-specific MAE; positive values indicate that the audience-specific predictions recover the audience labels more accurately. The values in brackets are 95% confidence intervals from a paired image-level bootstrap ($B = 2{,}000$) that resamples images within each tercile; the cost is a per-image difference (pooled minus audience-specific error on the same image).

[4]CLIP (Contrastive Language-Image Pretraining) is a pretrained vision-language model; we use its image embeddings as fixed visual representations (Radford et al., 2021).

[5]Intervals are 95% percentile confidence intervals from an image-level bootstrap ($B = 2{,}000$ resamples of the 138 test images).

Table 4 shows that the cost of using pooled predictions as a proxy for audience-specific evaluations increases with the magnitude of audience disagreement. Among images in the lowest-disagreement tercile, the two approaches perform similarly, and the pooled model produces slightly lower error (MAE $= 0.404$ versus $0.441$). The pattern reverses in the middle tercile, where pooling adds $0.09$ scale points of error, and becomes substantially larger in the highest-disagreement tercile, where MAE increases from $0.451$ under audience-specific prediction to $0.725$ under the pooled proxy. This supports Implication 1 that measurement loss increases with audience disagreement. At the same time, the overall average cost of $0.11$ masks substantial variation across images, with nearly all of the loss concentrated in the middle- and high-disagreement terciles, as predicted by Implication 2. The same pattern appears when disagreement is measured continuously (Appendix D).

Finally, we examine Implication 3: whether audience-specific structure appears for substantively meaningful partitions rather than arbitrary splits of respondents.

**Table 5: Audience-group divergence relative to a within-image permutation benchmark**

| Partition | Observed divergence | Null divergence | $z$ ($p_{\text{perm}}$) | ICC$(1,k)$ Group A | ICC$(1,k)$ Group B |
|---|---|---|---|---|---|
| Party (Dem. vs. Rep.) | 0.794 [0.760, 0.829] | 0.409 | 37 ($p < 0.001$) | 0.79 [0.77, 0.81] | 0.84 [0.83, 0.86] |
| Ideology (Lib. vs. Cons.) | 0.827 [0.790, 0.866] | 0.494 | 26 ($p < 0.001$) | 0.71 [0.69, 0.74] | 0.81 [0.79, 0.83] |

*Note:* Results use 960 immigration images, with at least three ratings per group and image. Divergence is the mean absolute difference between the two groups' image-level mean labels; bracketed values are 95% confidence intervals from an image-level bootstrap ($B = 2{,}000$ resamples of the images), reflecting resampling of these images rather than sampling of images from a population. The null column reports the mean divergence over 2,000 within-image permutations of group labels that preserve each image's observed group sizes, representing the divergence expected when group membership is unrelated to ratings. The $z$ statistic compares the observed divergence with the permutation mean and standard deviation; no permutation produced a divergence at least as large as the observed value, so $p_{\text{perm}} = 0.0005$. ICC$(1,k)$ reports the reliability of each group's image-level mean with 95% intervals. For party, Group A denotes Democrats and Group B Republicans; for ideology, Group A denotes liberals and Group B conservatives.

Partisanship and ideology produce substantially more between-group divergence than expected under random within-image relabeling, while the resulting group-specific image means remain reliably measured (Table 5). Thus, the observed audience differences

are not a mechanical consequence of dividing a finite set of ratings into two groups. They arise for substantively motivated partitions that identify systematic differences in image evaluations.

These results show that image-specific audience differences are reproduced across independent respondents and remain consequential when predicting sentiment for out-of-sample images. In this application, the measurement output should therefore preserve separate Democratic and Republican image evaluations rather than impose a single pooled label or one uniform partisan adjustment. When the research question instead targets a mixture of audiences, researchers can combine the audience-specific estimates using declared target weights $q$.

# 5 Consequences for Substantive Inference

Having used raw means in the prediction exercise to isolate the consequences of target aggregation, we return to the preferred model-based audience estimates for substantive inference. Substantive conclusions can depend on the estimand underlying an image-level sentiment measure. We illustrate this consequence by examining how independently extracted visual features (see Appendix E for construction details) are associated with Democratic sentiment, Republican sentiment, and pooled sentiment. The image-varying model produces separate Democratic and Republican image estimates. We use these as the audience-specific outcomes and combine them with the same audience weights used for the conventional pooled label to obtain a comparable model-based pooled outcome. We then estimate identical content regressions for each outcome and directly test whether the Democratic and Republican coefficients differ. This comparison shows whether aggregation preserves relationships shared across audiences, attenuates opposing relationships through cancellation, or masks that an association observed for the pooled label is concentrated primarily in one audience.

**Table 6: Consequences of aggregation for associations between visual content and sentiment**

| Content concept | Democratic $\hat{\theta}_{iD}$ | Republican $\hat{\theta}_{iR}$ | Dem.–Rep. difference | Pooled (raw) | Pooled (model-based) |
|---|---|---|---|---|---|
| American flag† | $-0.04^{*}$ | $0.12^{***}$ | $-0.17^{***}$ | $0.05^{*}$ | $0.04^{*}$ |
| | (0.018) | (0.024) | (0.024) | (0.022) | (0.018) |
| Law enforcement‡ | $-0.13^{***}$ | $-0.02$ | $-0.11^{***}$ | $-0.12^{***}$ | $-0.08^{***}$ |
| | (0.017) | (0.023) | (0.020) | (0.021) | (0.018) |
| Children & families‡ | $0.02$ | $-0.14^{***}$ | $0.16^{***}$ | $-0.07^{**}$ | $-0.06^{***}$ |
| | (0.018) | (0.022) | (0.019) | (0.022) | (0.018) |
| Protest‡ | $-0.01$ | $-0.17^{***}$ | $0.16^{***}$ | $-0.09^{***}$ | $-0.08^{***}$ |
| | (0.019) | (0.021) | (0.019) | (0.023) | (0.018) |
| Detention | $-0.13^{***}$ | $-0.28^{***}$ | $0.15^{***}$ | $-0.25^{***}$ | $-0.21^{***}$ |
| | (0.018) | (0.020) | (0.021) | (0.019) | (0.016) |
| Border barrier | $-0.15^{***}$ | $-0.13^{***}$ | $-0.03$ | $-0.17^{***}$ | $-0.14^{***}$ |
| | (0.016) | (0.025) | (0.019) | (0.023) | (0.019) |
| Crowd of migrants‡ | $-0.03$ | $-0.27^{***}$ | $0.24^{***}$ | $-0.17^{***}$ | $-0.15^{***}$ |
| | (0.018) | (0.020) | (0.020) | (0.021) | (0.017) |

*Note:* $N = 960$ images. Each cell reports the coefficient from a separate bivariate regression (with the image as the unit of observation) of an image-level sentiment label on a single standardized zero-shot CLIP image-content measure; the coefficient represents the change in sentiment associated with a one-standard-deviation increase in that content measure. The content measures are continuous similarity scores rather than mutually exclusive categories. The Democratic and Republican outcomes are the model-based audience-specific image estimates $\hat{\theta}_{iD}$ and $\hat{\theta}_{iR}$ from the image-varying measurement model (Table 3). The Dem.–Rep. column reports the difference between the two audience coefficients, estimated using an interaction model with standard errors clustered by image. Standard errors are reported in parentheses: HC1 heteroskedasticity-robust for the four audience and pooled columns, and clustered at the image level for the Dem.–Rep. difference. Because $\hat{\theta}_{iD}$ and $\hat{\theta}_{iR}$ are themselves estimated, these standard errors condition on the estimated labels and do not propagate the uncertainty in estimating them. The two pooled columns use the same audience-composition weights. The raw column pools the observed ratings, whereas the model-based column combines the two audience-specific model estimates. Their similarity shows that the pooling patterns are not produced by comparing model-based audience labels with a raw pooled label. † denotes opposite-sign audience associations; ‡ denotes an association concentrated primarily in one audience. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Table 6 shows the cancellation pattern for American flag imagery. A one-standard-deviation increase in flag content is associated with a $0.04$-point decrease in Democratic sentiment and a $0.12$-point increase in Republican sentiment. The direct test confirms that the two coefficients differ ($\widehat{\beta}_D - \widehat{\beta}_R = -0.17$, $p < 0.001$). The model-based pooled label, however, yields a small positive coefficient of $0.04$. Here aggregation combines associations of opposite sign into a weak pooled relationship, obscuring that flag imagery is associated with sentiment in different directions across the two audiences.

A second pattern appears when the pooled association is driven primarily by one audience. The children and families frame, for example, has essentially no association with Democratic sentiment ($0.02$) but is associated with substantially more negative Republican sentiment ($-0.14$). The direct difference between the two coefficients is statistically significant ($\widehat{\beta}_D - \widehat{\beta}_R = 0.16$, $p < 0.001$), yet the pooled label yields a negative coefficient of $-0.06$. A similar pattern happens for protest imagery, which has little association with Democratic sentiment ($-0.01$) but a considerably stronger negative association with Republican sentiment ($-0.17$). In these cases, the pooled coefficient correctly describes the aggregated label, but it can be misinterpreted as a relationship that applies similarly across audiences when it is concentrated primarily among Republican respondents.

These associations also show that image-specific audience heterogeneity is related to recurring visual content rather than being idiosyncratic to particular images. We interpret these patterns descriptively rather than causally: visual features co-occur within a nonrandomly constructed image corpus, so the coefficients identify associations with audience-specific evaluations rather than effects of particular visual elements.

The similarity between the raw and model-based pooled coefficients shows that these patterns are not produced by comparing hierarchical audience-specific labels with a noisier raw aggregate. In both cases, aggregation combines distinct audience-specific relationships into a single pooled association. Consequently, even a carefully estimated pooled label may obscure whether an observed relationship is shared across audiences, reflects

opposing associations that partially cancel, or is concentrated primarily in one audience. Maintaining the audience-specific estimates preserves this information while still allowing researchers to construct an aggregate when a pooled estimand is appropriate for the research question. For researchers who need a single summary, this is the practical payoff of retaining audience-specific estimates even when the final measure is pooled: the pooled coefficient is the same, but it comes with a reading of whether it describes both audiences or one.

# 6 Discussion

Existing approaches to sentiment analysis, including visual sentiment, often begin from a reasonable premise: people may disagree, but much of that disagreement reflects noise around a common judgment that can be reduced through aggregation. When this premise holds, averaging labels improves precision without changing the quantity being measured. Our results show, however, that this logic does not always apply to political imagery. When disagreement is stable, systematic, and tied to the audiences evaluating the same image, pooling does more than remove individual-level noise: it combines distinct audience-specific evaluations into a single quantity. That pooled quantity may be appropriate when the research question concerns the mixture itself, but it should not be treated as a perceiver-independent property of the image.

We examine this distinction using immigration imagery and a theoretically motivated audience division between Democratic and Republican respondents. We do not select this division because particular images are already known to be polarizing, but because partisanship provides a substantive reason to expect differences in how immigration imagery is evaluated. The analysis shows that Democratic and Republican labels are not simply noisier versions of the same image-level sentiment. Their differences vary systematically across images, and models that retain this variation recover audience-specific evaluations

more accurately. The consequences of pooling are also concentrated among the images on which the two audiences disagree most.

The practical implication lies in how researchers construct image-level labels, rather than in the specific predictive architecture they ultimately fit. Researchers should first specify whose evaluations a label is intended to represent and retain the annotator information needed to examine theoretically motivated audience differences. An average audience difference alone does not necessarily require separate audience-specific targets: if one audience differs from another by approximately the same amount across the image corpus, an audience-wide adjustment may be sufficient. When the size or direction of that difference varies systematically across images and is reproduced among independent respondents, however, the image-level targets should preserve that variation. Models can then use image content to predict the resulting audience-specific labels for unseen images. If the question instead targets an aggregate, researchers should combine those predictions using weights chosen for the target population rather than the composition of the annotation sample. Once image-specific audience variation has been removed through aggregation, higher predictive accuracy does not establish that the desired quantities have been recovered; it only shows that the resulting pooled quantity can be predicted more accurately.

As political science increasingly relies on automated measures of textual, visual, and audio content, the quality of those measures depends on whether the labels supplied to the model correspond to the concept of interest, and predictive performance cannot establish that. Audience differences in the interpretation of political imagery are not themselves surprising. The contribution of this study is to show what follows when those differences enter the annotation process: aggregation can produce a precise and predictable label while changing the evaluation that the label represents. This does not imply that every annotator characteristic should define a separate label. Rather, researchers should define the relevant audience or audience mixture substantively and preserve audience-

specific variation when it is systematic, image-specific, and reproducible. For visual sentiment analysis, and for computational measurement more broadly, the central question is therefore not only how accurately a label can be predicted, but whose judgment the prediction is meant to recover.

## Funding

This work was supported by the Florida State University College of Social Sciences and Public Policy to O.G.

## Acknowledgments

The authors thank the following individuals for their valuable feedback and suggestions: Quintin Beazer, Yusaku Horiuchi, Kosuke Imai, Holger Kern, Santiago Olivella, Carlisle Rainey, the participants in seminars and workshops at Florida State University and in the 2026 Society for Political Methodology meeting.

## Competing Interests

The authors declare none.

## Research with Human Subjects

This study has been approved by the Florida State University Institutional Review Board with study IDs: STUDY00004983 and STUDY00006310. Participants provided informed consent prior to participation.

## Data Availability Statement

Replication data and code will be made available in accordance with the journal's data-access and research-transparency requirements.

## Disclosure of AI tools usage

The authors used Claude Opus 4.8 (Anthropic) and GPT-5.5 (OpenAI) in June-July 2026 for copyediting (e.g., identifying typographical errors and suggesting wording improvements), to assist with coding tasks and in the preparation of the replication archive (including debugging and verifying code execution across computing environments). The authors reviewed all outputs and are responsible for all content.

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# Appendix

## A Simulation Design and Additional Results

The simulations examine estimator performance under four structures of audience disagreement. Each Monte Carlo simulation contains 80 images and 320 respondents, equally divided between Audiences A and B. Each respondent rates 15 images, producing 30 ratings per audience per image in a balanced incomplete design. We consider four data-generating processes: no systematic audience difference; a uniform audience difference across all images; an audience difference that varies across images; and a continuous-attitude condition in which the association between respondent attitudes and ratings varies across images. The first three conditions correspond to the three audience structures compared in the main text. Simulation parameters are calibrated to the scale and annotation density of the empirical application, with $\mathrm{SD}(\delta_i) = 0.65$ in the image-specific condition. We compare three nested model-based estimators that permit no audience difference, one uniform audience difference, or an audience difference that varies by image. We also include a raw pooled mean and separate raw audience means as benchmarks. Performance is evaluated against two targets: the Audience B target, $q = (0, 1)$, and an equally weighted target, $q = (1/2, 1/2)$.

Table A.1 shows that the additional flexibility of the image-varying model carries little cost when a simpler audience structure is correct. Under no audience difference, the no-difference and image-varying models produce similar RMSE ($0.187$ versus $0.195$); the same is true under a uniform difference for the uniform and image-varying models ($0.196$ versus $0.198$). When disagreement varies by image, however, the restrictive models recover the Audience B target less accurately and their uncertainty intervals cover the true value too infrequently. In the image-specific condition, the image-varying model reduces RMSE to $0.261$, compared with $0.439$ under the no-difference model and $0.395$ under the

**Table A.1: Estimator performance across simulated audience structures: additional results and interval coverage.**

| Estimator | No gap | Uniform gap | Image-specific heterogeneity | Attitude × image |
|---|---|---|---|---|
| *RMSE, Audience B target* $q = (0, 1)$ | | | | |
| No audience difference | 0.187 | 0.272 | 0.439 | 0.549 |
| Uniform audience difference | 0.193 | 0.196 | 0.395 | 0.451 |
| Image-varying audience difference | 0.195 | 0.198 | 0.261 | 0.266 |
| Raw pooled mean | 0.212 | 0.291 | 0.437 | 0.555 |
| Raw audience means | 0.300 | 0.307 | 0.301 | 0.307 |
| *RMSE, equally weighted target* $q = (1/2, 1/2)$ | | | | |
| No audience difference | 0.187 | 0.189 | 0.195 | 0.203 |
| Uniform audience difference | 0.187 | 0.189 | 0.195 | 0.202 |
| Image-varying audience difference | 0.187 | 0.189 | 0.190 | 0.195 |
| Raw pooled mean | 0.212 | 0.216 | 0.214 | 0.218 |
| Raw audience means | 0.212 | 0.216 | 0.214 | 0.218 |
| *95% interval coverage, Audience B target; nominal coverage = 0.95* | | | | |
| No audience difference | 0.97 | 0.88 | 0.67 | 0.57 |
| Uniform audience difference | 0.97 | 0.97 | 0.74 | 0.68 |
| Image-varying audience difference | 0.97 | 0.97 | 0.97 | 0.97 |
| Raw pooled mean | 0.95 | 0.84 | 0.67 | 0.56 |
| Raw audience means | 0.94 | 0.94 | 0.94 | 0.94 |

*Note:* Reported values are averages across 300 Monte Carlo replications with 80 images and 30 ratings per audience per image. The first three estimators impose no audience difference, one uniform difference, or an image-varying difference. The raw pooled mean ignores audience membership, while the raw audience means are calculated separately within each audience. The Audience B target assigns all weight to Audience B; the equally weighted target assigns one-half to each audience. The first two panels report RMSE, with lower values indicating more accurate recovery of the specified image-level target. The final panel reports the proportion of 95% intervals for the Audience B target that contain the true simulated value. Coverage near 0.95 indicates well-calibrated uncertainty; lower coverage indicates that the intervals miss the true target too frequently.

uniform-difference model, while coverage increases from $0.67$ and $0.74$ to $0.97$. The same pattern appears in the continuous-attitude condition: RMSE falls from $0.549$ and $0.451$ to $0.266$, and coverage rises from $0.57$ and $0.68$ to $0.97$. For the equally weighted target, differences among the model-based estimators are small (RMSE differs by no more than $0.008$) because the target itself averages the two audience evaluations. The raw audience means maintain approximately nominal coverage ($0.94$ in every condition) but have higher RMSE ($0.300$–$0.307$) because they are estimated separately for each image without the precision gains provided by the hierarchical model.

# B Data and Label Reliability

The empirical analysis uses 59,542 ratings of 960 immigration images from 5,533 Democratic and Republican respondents. Because respondents evaluated overlapping subsets of images rather than the full corpus, the number of ratings varies modestly across images. On average, each image received 31.4 Democratic ratings and 30.6 Republican ratings. We assess whether these ratings produce sufficiently reliable image-level audience means using intraclass correlation coefficients.

We estimate reliability using one-way random-effects models. ICC$(1, 1)$ measures the reliability of a single rating, whereas ICC$(1, k)$ measures the reliability of the image-level mean at the average effective number of raters per image. The effective group size is adjusted for the unbalanced number of ratings across images.

**Table B.1: Reliability of audience-specific image labels**

| Audience | Images | Avg. raters | ICC(1,1) [95% CI] | ICC$(1, k)$ [95% CI] |
|---|---|---|---|---|
| Democratic | 960 | 31.4 | 0.109 [0.098, 0.121] | 0.793 [0.774, 0.812] |
| Republican | 960 | 30.6 | 0.148 [0.135, 0.162] | 0.842 [0.827, 0.856] |
| All (pooled) | 960 | 62.0 | 0.095 [0.086, 0.104] | 0.867 [0.854, 0.879] |

*Note:* ICC(1,1) measures the reliability of a single rating, whereas ICC$(1, k)$ measures the reliability of the image-level mean at the average effective number of raters per image. Estimates are obtained from one-way random-effects models using classical ANOVA. Because the number of ratings varies across images, the average rater count is the bias-corrected effective group size for the unbalanced design.

Although individual ratings are noisy, the approximately 30 ratings collected per audience produce reliable audience-specific image means. ICC$(1, k)$ is 0.793 for Democratic labels and 0.842 for Republican labels. The pooled image mean has higher reliability, 0.867, because it contains approximately twice as many ratings per image. Its single-rating ICC is lower because, once Democratic and Republican ratings are pooled, systematic partisan differences are counted as disagreement among respondents rating the same image.

# C Attitude and Construct-Validity Analyses

A possible explanation for the Republican–Democratic differences is that partisanship primarily captures respondents' general attitudes toward immigration. We examine this possibility using the subset of survey waves that also asked respondents about their immigration attitudes. We compare a party-only model, an immigration attitude-only model, and a joint model containing both characteristics. Each model allows the corresponding association with sentiment to vary by image. In the joint model, $\delta_i$ denotes the image-specific partisan deviation and $\lambda_i$ denotes the image-specific deviation in the association between immigration attitudes and sentiment.

**Table C.1: Partisanship and immigration attitudes in image evaluations**

| Quantity | Estimate |
|---|---|
| *Average Republican–Democratic difference* | |
| Party-only model | $-0.38$ $[-0.46, -0.29]$ |
| Joint party-and-attitude model | $-0.22$ $[-0.30, -0.15]$ |
| Reduction in absolute party coefficient | 40% $[31\%, 54\%]$ |
| *Image-specific variation* | |
| Partisan SD, party-only model, $\mathrm{SD}(\delta_i)$ | 0.58 $[0.54, 0.62]$ |
| Partisan SD, joint model, $\mathrm{SD}(\delta_i)$ | 0.34 $[0.30, 0.38]$ |
| Attitude SD, joint model, $\mathrm{SD}(\lambda_i)$ | 0.28 $[0.26, 0.30]$ |
| *Cross-validation on out-of-sample respondents* | |
| Party-only model RMSE | 1.71 |
| Attitude-only model RMSE | 1.70 |
| Joint model RMSE | 1.69 |

*Note:* Results use 48,963 ratings of 609 immigration images from 4,124 respondents. Immigration attitudes are measured using a harmonized policy item concerning support for deporting unauthorized immigrants, reverse-coded so that higher values indicate more pro-immigration attitudes, and standardized before estimation. All models include random effects for respondents and images. The party-only model allows the Republican–Democratic difference to vary by image; the attitude-only model allows the association between immigration attitudes and ratings to vary by image; and the joint model includes both image-specific slopes. The reported reduction in the absolute magnitude of the party coefficient is descriptive and should not be interpreted as a causal mediation estimate. RMSE is calculated using cross-validation in which respondents used for evaluation are not the ones used for model estimation; lower values indicate better prediction of ratings from unseen respondents. 95% confidence intervals are reported in the brackets.

Table C.1 shows that immigration attitudes account for part, but not all, of the partisan pattern. Including the attitude measure reduces the average Republican–Democratic difference from $-0.38$ to $-0.22$, a reduction of approximately 40%, and reduces the standard

deviation of the image-specific partisan difference from 0.58 to 0.34. Substantial image-specific partisan variation remains after accounting for respondents' general immigration attitudes. The joint model also estimates image-specific variation in the association between immigration attitudes and sentiment, $\mathrm{SD}(\lambda_i) = 0.28$. RMSE for unseen respondents is nearly identical across the party-only, attitude-only, and joint models ($1.71$, $1.70$, and $1.69$, respectively), indicating that the principal difference among the specifications concerns how they represent systematic variation in evaluations rather than their overall ability to predict individual ratings.

# D Prediction Architecture and Robustness Checks

Table D.1 reports overall predictive performance. All three sentiment targets contain visual structure that can be recovered from image content. On the test set, the model explains 43% of the variation in Democratic labels, 57% in Republican labels, and 54% in pooled labels. Although the Republican target is the most predictable here, this comparison does not identify which label is substantively appropriate. Predictive performance indicates how well a target can be recovered from image content, not whether it corresponds to the audience whose evaluation the researcher intends to measure.

**Table D.1: Prediction performance across sentiment labels**

| | Label | Cross-validation | | | Test | |
|---|---|---|---|---|---|---|
| Target | SD | $R^2$ (SD) | RMSE | $r$ | $R^2$ | $r$ |
| Democratic | 0.72 | 0.411 (0.015) | 0.560 | 0.647 | 0.431 | 0.664 |
| Republican | 0.83 | 0.497 (0.044) | 0.579 | 0.708 | 0.567 | 0.764 |
| Pooled | 0.64 | 0.435 (0.027) | 0.482 | 0.660 | 0.542 | 0.747 |

*Note:* All models use frozen CLIP ViT-B/32 image embeddings with ridge regression. Model development uses five-fold cross-validation on 822 images, and final performance is evaluated on a test set of 138 images that was not used during model development. SD denotes the standard deviation of the target label.

The main prediction analysis uses frozen CLIP ViT-B/32 image embeddings with ridge regression. We use the 768-dimensional output of the CLIP vision encoder, before projection into the joint image–text embedding space. To evaluate this choice, we benchmark several image representations and prediction heads under the same training and evaluation procedure. Each model is trained separately to predict the Democratic, Republican, and pooled image-level sentiment labels. Model development uses the same five-fold cross-validation split of 822 images, and final performance is evaluated on the separate test set of 138 images.

Table D.2 shows that pretrained image representations substantially outperform models trained directly on the available images. CLIP ViT-B/32 produces the strongest and most consistent cross-validated performance across the three sentiment targets. Ridge regression and support vector regression perform similarly, including the same mean test

**Table D.2: Benchmarking prediction architectures across sentiment targets**

| | Cross-validated $R^2$ | | | | Test |
|---|---|---|---|---|---|
| Approach | Dem | Rep | Pooled | Mean $R^2$ (SD) | Mean $R^2$ |
| Mean baseline (no image) | -0.005 | -0.004 | -0.004 | -0.004 (0.003) | -0.003 |
| Basic CNN (from scratch, pixels) | -0.035 | 0.041 | -0.028 | -0.007 (0.061) | 0.030 |
| ResNet50V2 (frozen) + Ridge | 0.168 | 0.223 | 0.168 | 0.186 (0.058) | 0.124 |
| DenseNet169 (frozen) + Ridge | 0.157 | 0.338 | 0.232 | 0.242 (0.053) | 0.342 |
| **CLIP ViT-B/32 (frozen) + Ridge** | **0.411** | **0.497** | **0.435** | **0.448 (0.029)** | **0.514** |
| CLIP ViT-B/32 (frozen) + SVR | 0.402 | 0.497 | 0.429 | 0.443 (0.021) | 0.514 |
| CLIP ViT-B/32 (frozen) + Keras MLP | 0.135 | 0.369 | -0.020 | 0.161 (0.094) | 0.253 |

*Note:* Candidate architectures are benchmarked separately on the Democratic, Republican, and pooled image-level sentiment targets under a common evaluation protocol. Model development uses 822 images with predefined five-fold cross-validation for model fitting and hyperparameter selection. Final performance is evaluated on a separate test set of 138 images that is not used during model development. Cross-validated $R^2$ is reported separately for each target. Mean $R^2$ is the average across the three targets; the value in parentheses is the average of the three target-specific standard deviations across folds. Mean test $R^2$ is the average coefficient of determination across the three prediction targets. Pretrained models are used as frozen feature extractors, and only the prediction head is estimated. The specification retained for the main analysis is shown in bold. We retain CLIP ViT-B/32 with ridge regression because it performs consistently among the strongest architectures while using the simplest prediction head.

$R^2$ of 0.514. We retain CLIP ViT-B/32 with ridge regression because it achieves slightly higher mean cross-validated performance ($R^2 = 0.448$ versus $0.443$) while using the simpler prediction head.

To assess whether the main findings might depend on the selected image representation, we repeat the audience-specific and pooled prediction analyses using several alternative frozen backbones. These include larger CLIP models, self-supervised vision transformers, and ImageNet-pretrained convolutional neural networks. All representations are paired with ridge regression and evaluated using the same image split and prediction procedure as the main specification.

Table D.3 shows that the pooling-cost pattern is not specific to CLIP ViT-B/32. For every image backbone, pooling produces substantially greater prediction error for images that Democratic and Republican respondents place in different sentiment categories. Pooling costs range from 0.132 to 0.221 for polarizing images, compared with 0.003 to 0.062 for non-polarizing images. The increase in error when audience evaluations diverge therefore persists across image representations with different architectures and training objectives.

**Table D.3: Robustness of pooling costs across image backbones**

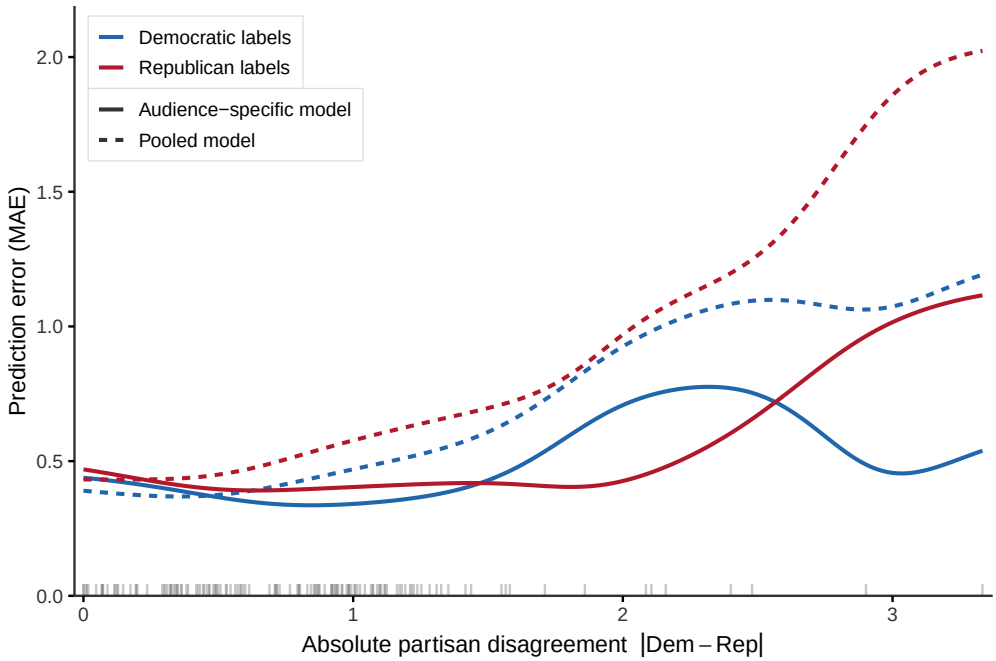
| Frozen backbone | Embedding dimension | Cost (non-polarizing) | Cost (polarizing) |
|---|---|---|---|
| **CLIP ViT-B/32 (OpenAI)** | **768** | **0.061** | **0.190** |
| CLIP ViT-L/14 (OpenAI) | 1024 | 0.059 | 0.221 |
| DINOv2 ViT-B/14 (self-supervised) | 768 | 0.049 | 0.177 |
| SigLIP ViT-B/16 (Google) | 768 | 0.042 | 0.219 |
| OpenCLIP ViT-B/32 (LAION-2B) | 512 | 0.062 | 0.181 |
| EVA-02-CLIP ViT-B/16 (merged-2B) | 512 | 0.047 | 0.201 |
| ResNet50V2 (ImageNet) | 2048 | 0.003 | 0.133 |
| DenseNet169 (ImageNet) | 1664 | 0.040 | 0.132 |

*Note:* All backbones are frozen, without fine-tuning, and paired with the same prediction head, ridge regression. Pooling cost is pooled-proxy MAE minus audience-specific MAE, evaluated on the separate test set and averaged across the Democratic and Republican targets. Positive values indicate that audience-specific predictions recover the corresponding audience labels more accurately. Sentiment categories are negative ($\leq 3$), neutral (3,5), and positive ($\geq 5$) on the 1–7 rating scale. An image is classified as polarizing when its Democratic and Republican mean labels fall in different categories and as non-polarizing when they fall in the same category. The backbone used in the main analysis is shown in bold.

We next disaggregate the continuous pooling-cost relationship by audience. Figure D.1 compares audience-specific and pooled-proxy prediction error for Democratic and Republican labels across the observed range of partisan disagreement.

For both audiences, pooled and audience-specific predictions perform similarly when Democratic and Republican evaluations are close, but the pooled model becomes less accurate as disagreement increases. The separation is larger for Republican labels in this application. Because relatively few test images have very high disagreement, the precise shape of the curves in the far-right tail should be interpreted cautiously.

## Figure D.1: Prediction error by audience and partisan disagreement




*Note:* Curves show kernel-smoothed mean absolute prediction error on the separate 138-image test set as a function of the absolute Democratic–Republican difference in mean sentiment ratings. Solid curves show audience-specific models evaluated against their corresponding audience labels; dashed curves show the pooled model evaluated separately against Democratic and Republican labels. Blue denotes Democratic labels and red denotes Republican labels.

# E Visual Content for the Inferential Analysis

To examine whether aggregation changes substantive conclusions about which visual features are associated with sentiment, we construct seven image-level visual content measures using zero-shot CLIP and relate them to the Democratic, Republican, and pooled sentiment labels reported in Table 6.

The sentiment outcomes are derived from the crowdsourced ratings and the image-varying hierarchical model described in the main text. The CLIP model is used only to measure the visual content of the images. The visual concepts and corresponding prompts were specified without reference to the observed Democratic–Republican differences in sentiment; they were not selected by examining which images produced the largest partisan disagreements. The resulting image-content measures are generated automatically from fixed text prompts and do not rely on manual image coding or on labeled data from this study used to train or fine-tune the model.

For the zero-shot content analysis, we use the `openai/clip-vit-base-patch32` model to project each image and text prompt into CLIP's 512-dimensional joint image–text space.[6] Image and text embeddings are normalized to unit length, so their dot product represents cosine similarity. Each visual concept is represented by several prompts, which are averaged to reduce sensitivity to any single wording. The prompt sets were suggested by Anthropic's Claude Opus 4.8 model and reviewed and finalized by the authors. For each image and concept, we subtract the image's mean similarity to the generic anchor prompts "a photo," "an image," and "a picture" from its mean similarity to the concept-specific prompts. We then standardize each resulting content score across the 960 images. Table E.1 reports the exact prompts used in the analysis.

For each visual-content measure, Table 6 estimates separate image-level regressions using the Democratic, Republican, raw pooled, and model-based pooled sentiment la-

[6]Unlike the prediction analysis, which uses the 768-dimensional vision-encoder representation, the zero-shot content measures use CLIP's 512-dimensional projected image–text embedding space so that image embeddings can be compared directly with text prompts.

**Table E.1: Text descriptions used to construct the visual-content measures**

| Visual concept | Text descriptions |
|---|---|
| Law enforcement | "a photo of police officers"; "immigration enforcement agents"; "a border patrol officer"; "ICE agents detaining someone" |
| Children & families | "a photo of immigrant children"; "a family with young children"; "a crying child"; "parents holding their kids" |
| Detention | "people held in a detention center"; "migrants behind a fence in custody"; "people in cages"; "a crowded holding facility" |
| Protest | "a protest demonstration with signs"; "people marching in a rally"; "activists holding banners" |
| Border barrier | "a border wall"; "a fence on the US-Mexico border"; "a border barrier" |
| American flag | "an American flag"; "patriotic American symbols"; "the stars and stripes" |
| Crowd of migrants | "a large crowd of migrants"; "a group of people crossing the border"; "a caravan of migrants" |

*Note:* The generic anchor descriptions are "a photo," "an image," and "a picture." The text descriptions were suggested by Anthropic's Claude Opus 4.8 model and reviewed and finalized by the authors.

bels. The Democratic and Republican outcomes are the model-based audience-specific image estimates from the image-varying hierarchical model. Differences between the Democratic and Republican coefficients are tested using a stacked interaction model with standard errors clustered by image.